

A redundant tight-frame erasure counterexample

Disproof of Conjecture 00000000952

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October 8, 2026

Scope. The conjecture bounds the worst spectral deviation after deleting k vectors from an n -dimensional uniform tight frame by k/n , with equiangular tight frames as equality cases. We use a redundant Parseval equiangular tight frame. Its original frame operator is exactly the identity, so the deviation is already normalized relative to the original tight-frame bound. No arbitrary scale factor is present.

The actual frame. In $H = \mathbb{R}^3$ take

$$v_0 = \frac{1}{2}(1, 1, 1)^T, \quad v_1 = \frac{1}{2}(1, -1, -1)^T, \quad v_2 = \frac{1}{2}(-1, 1, -1)^T, \quad v_3 = \frac{1}{2}(-1, -1, 1)^T.$$

Their inner products are

$$\langle v_i, v_j \rangle = \begin{cases} 3/4 & i = j, \\ -1/4 & i \neq j. \end{cases}$$

Thus their lengths are equal and the absolute pairwise inner products are equal: this is an equiangular family. Directly adding the four outer products gives

$$S = \sum_{i=0}^3 v_i v_i^T = I_3.$$

Indeed each diagonal entry is $4(1/4) = 1$, and each off-diagonal entry is a sum of two $1/4$ and two $-1/4$. Hence $\sum_i |\langle v_i, x \rangle|^2 = \langle Sx, x \rangle = \|x\|^2$. This proves that the four vectors are a uniform Parseval tight frame in dimension $n = 3$, with genuine redundancy and the ETF property. The same real matrices define a complex ETF in $\mathbb{C}^3$ if desired.

One erasure. Delete v_0 , so $k = 1$. The remaining frame operator and its deviation are

$$S_- = \sum_{i=1}^3 v_i v_i^T = I_3 - v_0 v_0^T, \quad D = S - S_- = v_0 v_0^T.$$

The nonzero vector v_0 is an eigenvector of D , with $Dv_0 = \|v_0\|^2 v_0 = (3/4)v_0$. Every vector orthogonal to v_0 lies in the kernel of D . Therefore the eigenvalues of S_- are $1/4, 1, 1$, and the worst spectral deviation from the original identity is

$$\|D\|_{\text{op}} = \frac{3}{4} > \frac{1}{3} = \frac{k}{n}.$$

This already occurs for an ETF, so neither the bound nor its asserted equality description holds as written. If uniform tight means unit-norm tight instead, multiply every vector by $2/\sqrt{3}$. The tight-frame bound becomes $4/3$, the erasure defect has norm 1, and the relative deviation remains $1/(4/3) = 3/4$. The counterexample therefore survives that usual normalization as well. Renormalizing after erasure or changing the denominator to the number of frame vectors would be a different statement from the given n -dimensional bound.

Formal verification. The Lean project uses complex-free real Euclidean space of dimension three and actual continuous linear frame operators, formed from $x \mapsto \langle v_i, x \rangle v_i$. It verifies all inner products, the full operator identity, the remaining operator and defect, and the eigenvector equation. Its final theorem derives $\|D\|_{\text{op}} \geq 3/4 > 1/3$ using the genuine operator norm. Thus the formal proof needs only a lower bound and does not assume the claimed norm value or replace the operator with a scalar certificate.